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| **2.3 Expressions and formulae** | understand that a letter may represent an unknown number or a variable<br>use correct notational conventions for algebraic expressions and formulae<br>substitute positive and negative integers, decimals and fractions for words and letters in expressions and formulae<br>use formulae from mathematics and other real-life contexts expressed initially in words or diagrammatic form and convert to letters and symbols | <br><br>Evaluate $2x - 3y$ when $x = -2$ and $y = 4$ |

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| **2.4 Linear equations** | solve linear equations, with integer or fractional coefficients, in one unknown in which the unknown appears on either side or both sides of the equation<br><br>set up simple linear equations from given data | $3x + 7 = 22$<br>$\frac{2}{3}x = 60$<br>$4x - 2 = 10 - x$<br>$5x + 17 = 3(x + 6)$<br>$\frac{15 - x}{4} = 2$<br>$\frac{1}{6}x + \frac{1}{3}x = 5$<br>The three angles of a triangle are $a°$, $(a + 10)°$, $(a + 20)°$.<br>Find the value of $a$ |
| **2.5 Proportion** | **Higher Tier only.** | |
| **2.6 Simultaneous linear equations** | calculate the exact solution of two simple simultaneous equations in two unknowns | $y = 2x$, $x + y = 12$<br>$x + y = 14$, $x - y = 2$ |
| **2.7 Quadratic equations** | **Higher Tier only.** | |
| **2.8 Inequalities** | understand and use the symbols $>$, $<$, $\geq$ and $\leq$<br><br>understand and use the convention for open and closed intervals on a number line<br><br>solve simple linear inequalities in one variable and represent the solution set on a number line<br><br>represent simple linear inequalities on rectangular cartesian graphs<br><br>identify regions on rectangular cartesian graphs defined by simple linear inequalities | To include double-ended inequalities eg $1 < x \leq 5$<br><br>$3x - 2 < 10$, so $x < 4$<br>$7 - x \leq 5$, so $2 \leq x$<br><br>Shade the region defined by the inequalities $x \geq 0$, $y \geq 1$, $x + y \leq 5$<br>Conventions for the inclusion of boundaries are not required |

| 3 Sequences, functions and graphs | | |
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| | **Students should be taught to:** | **Notes** |